

# TSX Trading Advisor

## Technical Reference Manual

*Architecture, Configuration, Data Flow, and AI Logic*

Classification: Confidential | Environment: Production | Platform: Windows / Python 3

### Table of Contents

1. System Architecture Overview
2. File Structure & Roles
3. Configuration Reference (config.json / .env)
4. Data Engine (market\_analyzer.py)
5. Technical Indicator Reference
6. AI Prompt System & Trading Rules
7. Telegram Daemon (telegram\_daemon.py)
8. Scheduled Cron Jobs & Automated Tasks
9. Trade Ledger (trades.csv)
10. Advice History System (advice\_history.txt)
11. Dependency Reference (requirements.txt)
12. Security Hardening
13. Operational Runbook
14. Troubleshooting Guide

## 1. System Architecture Overview

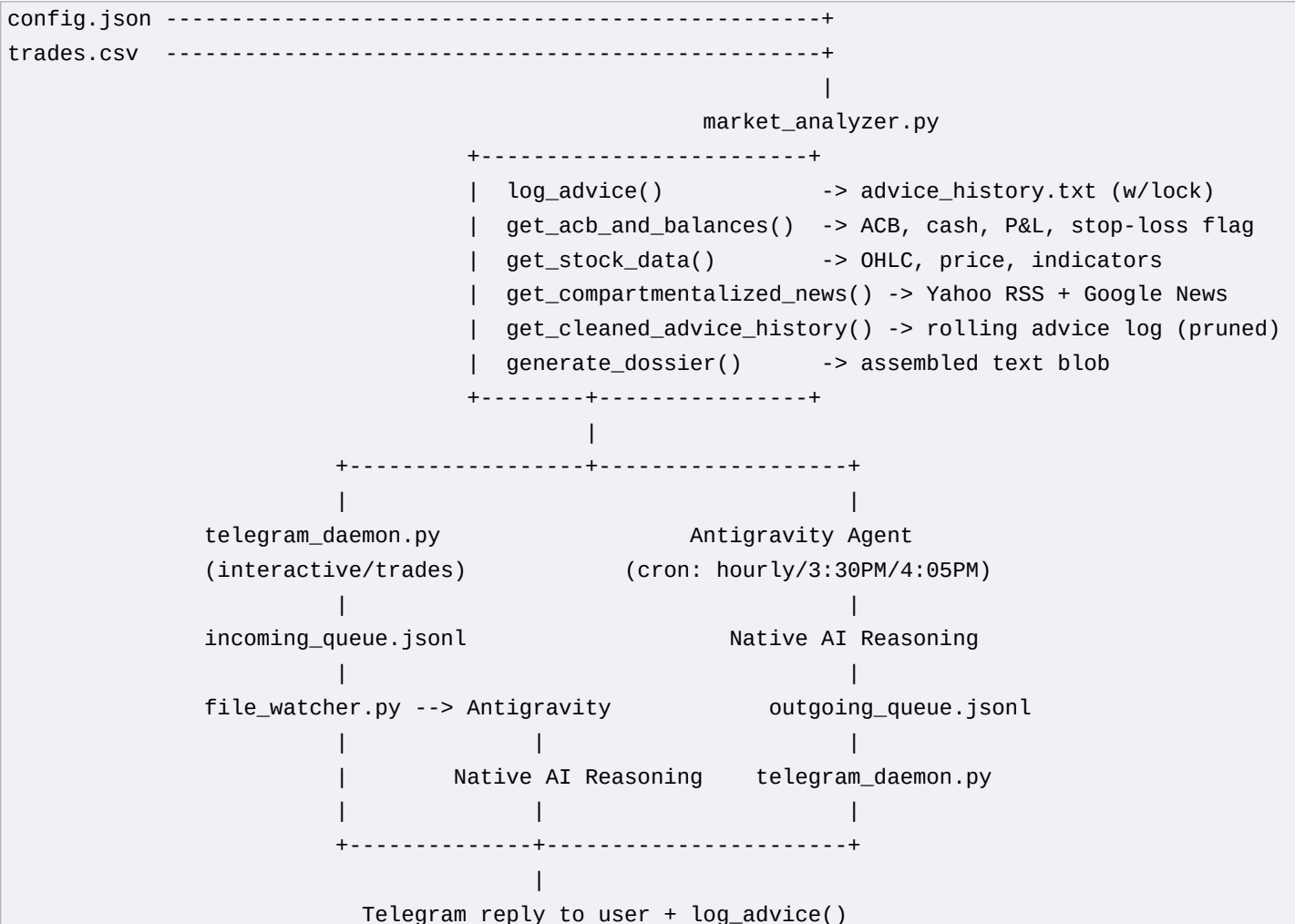
The TSX Trading Advisor is a locally-running, fully automated stock analysis and portfolio management system. It is built entirely on free, open-source Python libraries and requires no paid brokerage API. It connects to Yahoo Finance for real-time market data, fetches qualitative news from Yahoo Finance RSS and Google News RSS, and uses the Antigravity AI agent (running natively in the Antigravity session, no external API key required) to synthesize quantitative and qualitative signals into plain-language trading recommendations delivered via Telegram.

### 1.1 Core Execution Paths

The system runs along two independent, concurrent execution paths:

- PATH A - Telegram Daemon (telegram\_daemon.py): A long-running foreground process started manually via start\_daemon.bat. It listens for Telegram messages, handles natural-language trade logging, mode switching, on-demand analysis requests, and general Q&A. Messages are written to incoming\_queue.jsonl and read by the Antigravity agent via file\_watcher.py.
- PATH B - Antigravity Agent (Native Cron): The Antigravity AI agent runs three native cron tasks (hourly, 3:30 PM, 4:05 PM ET on weekdays). It generates a dossier directly by calling market\_analyzer.py and sends the resulting recommendation to Telegram by writing to outgoing\_queue.jsonl. No external LLM API key is needed; all AI reasoning is performed natively by the Antigravity agent.

### 1.2 Data Flow Diagram



### 1.3 Technology Stack

| Component            | Technology / Library                            | Version        |
|----------------------|-------------------------------------------------|----------------|
| Market Data          | yfinance (Yahoo Finance)                        | 1.4.1          |
| Technical Indicators | pandas-ta                                       | 0.4.71b0       |
| Data Frames          | pandas                                          | 3.0.1          |
| AI Reasoning         | Antigravity Agent (native, no API key required) | session-native |
| News (Qualitative)   | Yahoo Finance RSS + Google News RSS             | live           |
| Telegram Messaging   | python-telegram-bot                             | 22.8           |
| Secrets Management   | python-dotenv                                   | 1.2.2          |
| HTTP Requests        | requests                                        | latest         |
| PDF Generation       | fpdf2                                           | 2.8.7          |
| Message Bridge       | file_watcher.py (incoming_queue.jsonl)          | custom         |
| Runtime              | Python (Miniforge)                              | 3.x            |

## 2. File Structure & Roles

| File                     | Role                                                                                                     |
|--------------------------|----------------------------------------------------------------------------------------------------------|
| market_analyzer.py       | Core data engine. Computes indicators, fetches news from Yahoo RSS + Google News, builds the dossier     |
| telegram_daemon.py       | Long-running bot server. Listens for Telegram messages; writes them to incoming_queue.jsonl and sends    |
| file_watcher.py          | Lightweight bridge. Polls incoming_queue.jsonl and prints new messages to stdout to wake up the Antigrav |
| send_telegram.py         | Standalone CLI utility. Sends a text string via Telegram Bot API and logs advice to advice_history.txt.  |
| generate_pdf.py          | Generates this PDF documentation from source.                                                            |
| config.json              | Runtime configuration. Controls active mode (SHORT/MEDIUM), portfolio-to-news-topic mapping (includin    |
| trades.csv               | Source of truth for all holdings. Records every trade; used to calculate ACB and cash balance.           |
| advice_history.txt       | Rolling log of AI-generated advice. Injected into every new dossier for context continuity.              |
| incoming_queue.jsonl     | JSONL file written by telegram_daemon.py. Each line is a user message from Telegram.                     |
| outgoing_queue.jsonl     | JSONL file written by Antigravity agent. Messages here are sent to Telegram by telegram_daemon.py.       |
| .env                     | Secrets file. Contains TELEGRAM_BOT_TOKEN and TELEGRAM_CHAT_ID only.                                     |
| .gitignore               | Excludes .env, trades.csv, advice_history.txt, and *.lock files from version control.                    |
| start_daemon.bat         | Windows batch file to launch telegram_daemon.py from the correct working directory.                      |
| requirements.txt         | Pinned Python dependency list for reproducible installs.                                                 |
| AGENT_RESTORE_INSTRUCTIO | Instructions for an AI agent to restore the system if the Antigravity session is lost.                   |

### 3. Configuration Reference

#### 3.1 config.json

This file controls the operating mode and portfolio composition. It is read at runtime by `market_analyzer.py` on every dossier generation. The user can modify the mode via Telegram by sending "Short Term Mode" or "Medium Term Mode".

```
{
  "mode": "SHORT",          // "SHORT" or "MEDIUM"
  "portfolio": {
    "CNQ.TO": ["CNQ.TO", "CL=F", "canadian+oil+sector", "OPEC+oil+production"],
    "ABX.TO": ["ABX.TO", "GC=F", "gold+mining+sector", "federal+reserve+interest+rate"]
  },
  "stop_loss_pct": 5.0      // % drop below ACB that triggers a mandatory SELL directive
}
```

- "mode": Controls which technical indicators are computed and which AI system instructions are used.
- "portfolio": Keys are TSX ticker symbols. Values are lists of news search topics. Valid Yahoo Finance ticker symbols (e.g. CL=F, GC=F) are also fetched via Yahoo RSS and their live price/daily change is injected into the Macro Commodities dossier section. Plain text search terms (e.g. "canadian+oil+sector") are fetched from Google News RSS only and contribute qualitative headlines without generating price data.
- "stop\_loss\_pct": The maximum acceptable loss percentage before the system injects a hard STOP-LOSS TRIGGERED directive into the dossier. Default is 5.0%.
- To add a new stock, add a new key-value pair to "portfolio". The system will automatically fetch data and news for it on the next run. No code changes required.

#### 3.2 .env (Secrets)

All credentials are stored in a `.env` file in the project root. This file is excluded from version control by `.gitignore`. Never commit this file. Note: No Gemini or LLM API key is required -- all AI reasoning is performed natively by the Antigravity agent session.

```
TELEGRAM_BOT_TOKEN=xxxxxxxx:xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
TELEGRAM_CHAT_ID=xxxxxxxx # Used by BOTH send_telegram.py AND telegram_daemon.py
```

- TELEGRAM\_BOT\_TOKEN: BotFather API token. Required by both the daemon and `send_telegram.py`.
- TELEGRAM\_CHAT\_ID: Your numeric Telegram chat ID. Used by `send_telegram.py` to push outbound messages, and by `telegram_daemon.py` to authorize inbound messages. A single key serves both purposes.

#### 3.3 Changing Operating Mode

Mode can be changed at runtime without restarting the daemon by sending a message to the Telegram bot.

```
User sends: "change mode"
Bot replies: "Please reply with Short Term Mode or Medium Term Mode"
User sends: "Short Term Mode"
Bot replies: "Operating mode successfully updated to: SHORT TERM"
```

## 4. Data Engine (market\_analyzer.py)

market\_analyzer.py is the central data and prompt generation engine. All other components import from it. It now loads config.json exactly ONCE per generate\_dossier() call and passes the parsed config object to all sub-functions, eliminating any risk of a mid-generation config race condition.

### 4.1 Function Reference

#### **log\_advice(text)**

Centralized writer for advice\_history.txt. Acquires a filesystem lockfile (advice.lock) before writing, retrying up to 50 times (5s total) if the lock is already held by another process. Releases the lock in a finally block. Used by both send\_telegram.py and telegram\_daemon.py.

#### **get\_current\_mode(config=None)**

Returns "SHORT" or "MEDIUM". Accepts an optional pre-loaded config dict to avoid a redundant file read. Falls back to reading config.json directly if no config is passed.

#### **get\_tickers(config=None)**

Returns a list of ticker symbols from the portfolio keys. Accepts an optional pre-loaded config dict. Falls back to ["CNQ.TO", "ABX.TO"] if the file is absent.

#### **get\_stop\_loss\_pct(config=None)**

Returns the stop\_loss\_pct value from config.json (default 5.0). Accepts an optional pre-loaded config dict.

#### **get\_acb\_and\_balances()**

Reads trades.csv row by row and computes: (1) Adjusted Cost Base (ACB) per ticker. (2) Available Cash from the last Cash\_Balance column value. Malformed rows with non-numeric values are silently skipped via try/except ValueError. Returns: {"positions": {"CNQ.TO": {"shares": 100.0, "total\_cost": 6012.0, "acb": 60.12}}, "cash": 5230.00}

#### **get\_stock\_data(ticker\_symbol, mode)**

Fetches 6 months of OHLCV history via yfinance (timeout=10s). Now returns Open, High, and Low in addition to Close, for all modes. Computes technical indicators via pandas-ta. Wraps everything in try/except; returns an error string if the ticker is invalid.

#### **get\_compartmentalized\_news(config=None)**

Reads portfolio from config (or config.json directly), builds a unique list of news topics, then for each topic fetches headlines from BOTH Yahoo Finance RSS and Google News RSS (timeout=10s each). Headlines older than 48 hours are filtered out. Duplicates across sources are eliminated via a seen\_titles set. Each item is tagged with a freshness label (NEW/RECENT/OLD) based on age, and auto-categorized as [MACRO], [EARNINGS], [ANALYST], or [NEWS] via keyword matching. The <description> field is extracted and truncated to 200 characters to provide the AI with article summaries. Results are sorted by recency and capped at 10 items per topic.

#### **get\_cleaned\_advice\_history()**

Reads advice\_history.txt, parses entries by [YYYY-MM-DD HH:MM:SS] timestamp headers, and applies two separate policies: (1) PHYSICAL FILE: Rewrites the file retaining ALL entries for today + the last entry for each of the last 5 business days. Weekend entries are silently discarded. (2) LLM CONTEXT STRING: Returns only the LAST 5 entries from today + the last entry from each prior day. This keeps the LLM prompt token-efficient while the local log remains complete.

#### **generate\_dossier()**

The main assembler. Loads config.json once, then calls all sub-functions passing the config object. Sections: Unrealised

P&L per holding, deterministic STOP-LOSS TRIGGERED directives, Today Open/High/Low per ticker, a Macro Commodities section with live price and daily change for recognized ticker symbols (e.g. CL=F, GC=F), and the new intelligent qualitative news section with freshness scores, categories, and summaries from both Yahoo and Google News sources.

**get\_analysis\_prompt(dossier=None, mode=None)**

Wraps the dossier in the appropriate system instructions and output format rules for either SHORT or MEDIUM mode. Used by the Antigravity agent for all analysis tasks: scheduled cron tasks, on-demand requests from Telegram, and general Q&A.

**get\_eod\_summary\_prompt(dossier=None)**

Wraps the dossier in special end-of-day instructions that suppress active trading advice and instead request a narrative summary of portfolio performance and next-day outlook. Used exclusively by the 4:05 PM cron task.

## 5. Technical Indicator Reference

### 5.1 SHORT Mode Indicators (mode = "SHORT")

Designed for a 5-to-7 day momentum trading horizon. Optimized for mean-reversion signals.

| Indicator                        | Parameters         | Interpretation                                            |
|----------------------------------|--------------------|-----------------------------------------------------------|
| EMA (Exponential Moving Average) | 5-day, 9-day       | Short-term momentum direction. EMA_5 crossing above       |
| RSI (Relative Strength Index)    | 7-day              | Overbought >70, Oversold <30. Faster than standard 14     |
| ATR (Average True Range)         | 5-day              | Measures raw volatility. Used to set stop-loss distances. |
| Bollinger Bands                  | 20-day, 2 std devs | Price below BB_Lower = oversold/compressed. Price ab      |
| Vol_Ratio                        | 10-day rolling avg | Volume / 10-day average. >2.0 during a drop signals cap   |

### 5.2 MEDIUM Mode Indicators (mode = "MEDIUM")

Designed for a 2-to-4 week swing trading horizon. Emphasizes mean price and income events.

| Indicator                     | Parameters         | Interpretation                                            |
|-------------------------------|--------------------|-----------------------------------------------------------|
| RSI (Relative Strength Index) | 14-day             | Standard 14-day RSI. Overbought >70, Oversold <30.        |
| SMA (Simple Moving Average)   | 50-day             | Key support/resistance level. Price above SMA_50 is str   |
| Bollinger Bands (Upper)       | 50-day, 2 std devs | Profit-taking trigger. AI instructed to recommend trimmin |
| Ex-Dividend Date              | from ticker.info   | Upcoming ex-div dates affect entry timing. Holding befor  |

### 5.3 ACB (Adjusted Cost Base) Calculation

The system computes ACB from scratch on every run using a pure roll-forward method over trades.csv. It does NOT rely on the stored Shares\_Balance or Cash\_Balance columns for ACB computation (only cash reads from the column, for simplicity). The ACB formula:

```
On BUY or INIT:
    shares += quantity
    total_cost += quantity * price
    acb = total_cost / shares

On SELL:
    total_cost -= quantity * acb # Reduce cost at current ACB, not sell price
    shares -= quantity
    # If shares drop to 0, reset total_cost and acb to 0.0
```



## 6. AI Prompt System & Trading Rules

The AI prompt system is the decision-making core of the advisor. The system instructions embedded in the prompts constitute a formal rule set that the Gemini model must follow. These rules are different depending on the active mode.

### 6.1 Dossier Structure (Input to LLM)

The dossier is a structured plain-text document assembled by `generate_dossier()`. It contains:

- Section 1: Current Mode header (SHORT or MEDIUM)
- Section 2: Date and timestamp
- Section 3: Portfolio & Pricing -- shares owned, ACB, current price, Today Open/High/Low, Unrealised P&L, any STOP-LOSS TRIGGERED directives, and all technical indicators per ticker
- Section 4: Macro Commodities -- live price and daily % change (vs Open) for recognized ticker symbols configured as news topics (e.g. CL=F for oil, GC=F for gold)
- Section 5: Qualitative Data -- up to 10 recent headlines per configured news topic, sourced from both Yahoo Finance RSS and Google News RSS. Each headline is tagged with a freshness label (NEW/RECENT/OLD), age in hours, article category [MACRO/EARNINGS/ANALYST/NEWS], and a truncated article summary. Headlines older than 48 hours are excluded.
- Section 6: Previous Advice History -- last 5 entries from today + final entry from each of the last 5 business days (weekend entries discarded)

### 6.2 SHORT Mode System Instructions (all 10 rules)

| Rule # | Name                                 | Description                                                                      |
|--------|--------------------------------------|----------------------------------------------------------------------------------|
| 1      | Trend Validation (No Falling Knives) | Do not assume a snap-back rally if $EMA_5 < EMA_9$ . Wait 30 minutes.            |
| 2      | Volume Shock / Capitulation          | $Vol\_Ratio > 2.0$ during a price drop = panic selling exhaustion.               |
| 3      | Market Open Volatility               | Before 9:30 AM ET, do NOT recommend buying. Wait 30 minutes.                     |
| 4      | Finalized Macro Events               | If news confirms a deal is done, anticipate the asset movement.                  |
| 5      | Cash Constraint                      | If free cash < cost of purchase, BUY is strictly forbidden.                      |
| 6      | Position Sizing (Tranching)          | Recommend buying in tranches (e.g., 25% of free cash).                           |
| 7      | Protective Stop-Loss for Holds       | Explicitly recommend to SELL if price drops more than 10% from entry.            |
| 8      | Profit-Taking Mechanics              | Price at $BB\_Upper$ AND $RSI > 70$ = recommend trimming.                        |
| 9      | Zero-Share Holdings                  | If 0 shares owned and not recommending BUY, use AVOID.                           |
| 10     | Acknowledge Failed Theses            | If previous predictions failed and the stock continues dropping, recommend SELL. |

### 6.3 MEDIUM Mode System Instructions (6 rules)

Medium mode uses rules 5-10 from above (Cash Constraint, Tranching, Protective Stop-Loss, Profit-Taking, Zero-Share, Failed Theses). Rules 1-4 are omitted because Bollinger-based mean-reversion and volume shock are short-term signals not relevant to a multi-week swing trading thesis. The  $SMA_{50}$  and  $RSI_{14}$  take their place as primary indicators.

### 6.4 AI Output Format

Both modes require the LLM to structure its response in exactly three sections:

- Section 1: A brief 1-2 sentence introduction summarizing the market environment.
- Section 2: A numbered list of priority actions (BUY, SELL, HOLD). AVOID and IGNORE are never listed here.

- Section 3: Stock-specific analysis for actioned stocks only. Maximum 3 sentences per stock. Indicators integrated naturally into prose (not as bullet points).

### 6.5 EOD Summary Prompt (4:05 PM Only)

`get_eod_summary_prompt()` uses a different instruction set that suppresses active trading advice. The AI is instructed to produce a 3-section response: (1) End-of-Day market summary, (2) Portfolio performance for currently held positions only, (3) 2-3 sentence outlook for tomorrow based on today's price action, volume, and breaking news.

### 6.6 Two-Step Intent Detection (Q&A Optimization)

When a user sends a free-form message to the Telegram bot (not "analyze" or a trade), the daemon runs a two-step process to avoid wasting tokens:

- Step 1 (Lightweight): A micro-prompt is sent to Gemini asking only whether the message is a "TRADE" or a "QUESTION". The prompt requests a JSON response. No dossier is included.
- Step 2A (TRADE path): If intent = TRADE, the extracted JSON (action, ticker, quantity, price) is parsed and staged in memory as a `pending_trade`, awaiting YES/NO confirmation.
- Step 2B (QUESTION path): If intent = QUESTION, the full dossier is generated, appended to a Q&A prompt, and sent to Gemini for a detailed response. This is the expensive path.

## 7. Telegram Daemon (telegram\_daemon.py)

The Telegram daemon is a persistent Python process built on the python-telegram-bot library. It uses long-polling (not webhooks) to receive messages. It must be started manually and runs in the foreground.

### 7.1 Message Routing Logic

Incoming messages are routed through the following priority-ordered decision tree:

```
Incoming message
|
+-- Is chat_id authorized? (matches TELEGRAM_CHAT_ID in .env)
|   NO  -> Silently drop message (print to console only)
|   YES -> Continue
|
+-- Is there a pending_trade for this chat_id AND message is "yes"/"no"?
|   YES -> Confirm or cancel staged trade -> update_ledger() -> done
|
+-- Message is "change mode"?
|   YES -> Ask user to specify Short or Medium term mode
|
+-- Message is "short term mode" or "medium term mode"?
|   YES -> Write new mode to config.json -> done
|
+-- Message is "analyze" or "analysis"?
|   YES -> get_analysis_prompt() -> Gemini -> send_chunked_reply() -> log_advice()
|
+-- Everything else -> Two-step intent extraction
    TRADE intent    -> stage in pending_trades -> ask for confirmation
    QUESTION intent -> generate_dossier() -> Gemini Q&A -> reply
```

#### 7.1a Available Commands

| Command / Message                    | Response                                                            |
|--------------------------------------|---------------------------------------------------------------------|
| /start                               | Welcome message with brief instructions.                            |
| /help                                | Full capability list: trade logging, analyze, change mode, Q&A.     |
| analyze (or analysis)                | Generates a full live market analysis with trading recommendations. |
| change mode                          | Prompts to switch between Short Term and Medium Term modes.         |
| Short Term Mode                      | Sets mode to SHORT in config.json immediately.                      |
| Medium Term Mode                     | Sets mode to MEDIUM in config.json immediately.                     |
| Natural trade (e.g. I bought 100 CN) | Two-step confirm-then-commit trade logging flow.                    |
| Any question                         | Generates a live-dossier-backed Q&A response from Gemini.           |

### 7.2 Trade Logging Flow

Trades go through a two-phase confirm-then-commit process to prevent accidental ledger writes:

- Phase 1 (Stage): Gemini extracts action, ticker, quantity, and price from the message. The trade dict is stored in the in-memory pending\_trades{chat\_id} dict.
- Phase 2 (Commit): User replies YES. The daemon calls update\_ledger(action, ticker, qty, price). This function calls get\_acb\_and\_balances() for current state, validates cash/share sufficiency, and appends a new row to

trades.csv.

- Limitation: pending\_trades is in-memory only. If the daemon restarts between Phase 1 and Phase 2, the staged trade is lost and must be re-entered.

### 7.3 update\_ledger() Function

This function writes new trade rows to trades.csv. It validates before writing:

- BUY validation: if cash\_bal < cost, raises ValueError("Insufficient cash!")
- SELL validation: if current\_shares < quantity, raises ValueError("Insufficient shares")
- Appends a CSV row: [Date, Ticker, Action, Quantity, Price, Shares\_Balance, Cash\_Balance]

### 7.4 Market Hours Warning

When the user sends "analyze" outside of trading hours (9:30 AM to 4:00 PM ET, Mon-Fri), the response is prefixed with a warning: "The market is currently closed. This analysis is based on the last session's closing data." Note: the time check uses local system time and assumes the machine is in the ET timezone.

### 7.5 Message Chunking

Telegram enforces a hard 4096-character limit per message. The send\_chunked\_reply() async function splits any response into 4000-character chunks and sends them sequentially to avoid truncation.

## 8. Scheduled Cron Jobs & Automated Tasks

The three scheduled tasks are managed by the Antigravity AI agent using its native schedule tool. When a task fires, the Antigravity agent directly executes a Python script to call `market_analyzer.generate_dossier()`, then performs all reasoning natively (no external LLM API required), and delivers the recommendation by writing to `outgoing_queue.jsonl`, which the Telegram daemon reads and sends to the user. The advice is also logged to `advice_history.txt` via `log_advice()` for context continuity.

### 8.1 Task 1: Hourly Market Scan

| Parameter       | Value                                                         |
|-----------------|---------------------------------------------------------------|
| Cron Expression | 0 10-15 * * 1-5                                               |
| Schedule        | Every hour on the hour, 10 AM to 3 PM ET, Monday to Friday    |
| Reasoning       | Antigravity agent native AI (no API key)                      |
| Output          | SHORT or MEDIUM mode trading recommendation sent via Telegram |

### 8.2 Task 2: Pre-Close Market Scan (3:30 PM)

| Parameter       | Value                                                                                                                    |
|-----------------|--------------------------------------------------------------------------------------------------------------------------|
| Cron Expression | 30 15 * * 1-5                                                                                                            |
| Schedule        | Every weekday at 3:30 PM ET                                                                                              |
| Reasoning       | Antigravity agent native AI (no API key)                                                                                 |
| Output          | Final intraday recommendation before market close. The last advice logged to <code>advice_history.txt</code> for the day |

### 8.3 Task 3: End-of-Day Summary (4:05 PM)

| Parameter       | Value                                                                         |
|-----------------|-------------------------------------------------------------------------------|
| Cron Expression | 5 16 * * 1-5                                                                  |
| Schedule        | Every weekday at 4:05 PM ET (5 minutes after market close)                    |
| Reasoning       | Antigravity agent native AI (EOD summary mode, no API key)                    |
| Output          | Holistic EOD summary: portfolio performance for the day and next-day outlook. |

### 8.4 Advice History: Intraday vs File Retention Policy

Throughout the day, every recommendation generated by the cron tasks or `/analyze` command is appended to `advice_history.txt` via the centralized `log_advice()` function. When `get_cleaned_advice_history()` runs (on every dossier generation), it applies a two-layer retention policy:

- PHYSICAL FILE retention: ALL entries for today are preserved. Only the last entry from each of the past 5 business days is kept. Weekend entries are silently discarded.
- LLM CONTEXT passed to Gemini: Only the LAST 5 entries from today (not all of them). This keeps the prompt token-efficient while the local log remains complete for review.
- History limit: Maximum 5 previous business days. Older entries are permanently deleted from the file.

## 9. Trade Ledger (trades.csv)

trades.csv is the single source of truth for all financial positions. It is an append-only CSV file. The daemon appends new rows via `update_ledger()`. Manual edits are possible but must preserve the column schema exactly.

### 9.1 Schema

| Column         | Type                | Description                                                                                          |
|----------------|---------------------|------------------------------------------------------------------------------------------------------|
| Date           | YYYY-MM-DD HH:MM:SS | Timestamp of the trade (local system time).                                                          |
| Ticker         | String              | TSX ticker symbol, e.g., "CNQ.TO". Always uppercase.                                                 |
| Action         | String              | "BUY", "SELL", or "INIT" (initial portfolio seeding). INIT is used to seed the initial cash balance. |
| Quantity       | Float               | Number of shares traded.                                                                             |
| Price          | Float               | Per-share price at time of trade.                                                                    |
| Shares_Balance | Float               | Running shares balance for this ticker after the trade. Should be 0 for INIT.                        |
| Cash_Balance   | Float               | Total available free cash after the trade. This is the canonical cash figure.                        |

### 9.2 Example

```
Date,Ticker,Action,Quantity,Price,Shares_Balance,Cash_Balance
2026-06-14 09:00:00,INIT,INIT,0,0,0,20000.00
2026-06-14 10:30:00,CNQ.TO,BUY,100,62.50,100.0,13750.00
2026-06-15 14:00:00,CNQ.TO,BUY,50,60.12,150.0,10756.00
2026-06-16 11:00:00,CNQ.TO,SELL,30,64.00,120.0,12676.00
```

### 9.3 Important Notes

- INIT action: The very first row is typically an INIT action used to seed the initial cash balance. The Ticker column value for INIT rows is ignored by the ACB calculation.
- Cash\_Balance is the canonical cash figure: `get_acb_and_balances()` reads the last Cash\_Balance value from the file. If this column is ever manually set incorrectly, the cash constraint logic will be wrong.
- The file uses UTF-8 encoding and should use consistent line endings (LF or CRLF). Python's csv module with `newline=""` handles this transparently on append.

## 10. Advice History System (advice\_history.txt)

advice\_history.txt provides the AI with rolling memory of its own recommendations. Without it, the AI would have no context about what it said earlier in the day and could give contradictory advice. The file uses a simple timestamp-keyed format.

### 10.1 File Format

```
[2026-06-16 10:00:15]
The TSX market opened with moderate volatility...
1. HOLD CNQ.TO

[2026-06-16 11:00:22]
Oil prices have recovered slightly...
1. HOLD CNQ.TO

[2026-06-15 15:30:07]
(Final advice from previous business day)
1. BUY CNQ.TO
```

### 10.2 Writers

- `log_advice(text)` in `market_analyzer.py`: The single canonical writer. Both `send_telegram.py` and `telegram_daemon.py` import and call this function. It acquires a filesystem lockfile (`advice.lock`) before writing to prevent concurrent write corruption.

### 10.3 Pruning (`get_cleaned_advice_history`)

On every call, the function: (1) Parses all timestamp blocks via regex. (2) Groups entries by calendar date. (3) Rewrites the physical file retaining ALL entries for today and only the last entry per day for the last 5 business days. (4) Returns a bounded LLM context string with only the last 5 entries from today (not all) plus the last prior-day entry per retained date.

### 10.4 Concurrency Safety

All writes to `advice_history.txt` now route through `log_advice()`, which uses OS-level lockfile semantics (`os.O_CREAT | os.O_EXCL`) to guarantee exclusive write access. The lock is released in a finally block to prevent stale locks. In the event of a very long lock wait (>5 seconds), the write proceeds anyway to prevent advice from being permanently lost.

## 11. Dependency Reference (requirements.txt)

| Package             | Pinned Version | Purpose                                                    |
|---------------------|----------------|------------------------------------------------------------|
| yfinance            | 1.4.1          | Fetches OHLCV price history and ticker metadata (ex-di     |
| pandas              | 3.0.1          | DataFrame manipulation for price history and indicator c   |
| pandas-ta           | 0.4.71b0       | Technical Analysis library. Computes EMA, RSI, ATR, S      |
| python-telegram-bot | 22.8           | Async Telegram Bot API wrapper. Powers the daemon's        |
| python-dotenv       | 1.2.2          | Reads the .env file and injects variables into os.environ. |
| requests            | latest         | HTTP client. Used by send_telegram.py to call the Teleg    |
| fpdf2               | 2.8.7          | PDF generation library. Used exclusively by generate_p     |

### 11.1 Installation

```
# Install all dependencies:
C:\Users\Aamir\miniforge3\python.exe -m pip install -r requirements.txt
```



## 12. Security Hardening

### 12.1 Telegram Bot Authorization

The daemon compares every incoming message's `chat_id` against the `TELEGRAM_CHAT_ID` value in `.env`. If they do not match, the message is silently dropped (no reply) and a warning is logged to the console. This prevents unauthorized users from logging trades or extracting portfolio data even if they discover the bot token.

```
# In handle_message():
if str(chat_id) != os.getenv("TELEGRAM_CHAT_ID"):
    print(f"Unauthorized access attempt from {chat_id}")
    return
```

### 12.2 Secrets Management

- `.env` is excluded from version control by `.gitignore`.
- `trades.csv` and `advice_history.txt` are also excluded, as they contain sensitive financial data.
- No credentials are hardcoded in any `.py` file. All secrets are read from `os.getenv()`.

### 12.3 Input Validation

- Trade quantity and price are parsed by the Gemini model and then cast to `float()` with explicit error handling. A `ValueError` terminates processing safely.
- Ticker validation: The intent-extraction prompt is given the list of valid tickers from `config.json`, instructing the model to only extract trades for configured symbols.
- Cash and share sufficiency checks in `update_ledger()` prevent the ledger from entering a negative balance state.

## 13. Operational Runbook

### 13.1 Daily Startup

Step 1: Open File Explorer and navigate to c:\Users\Aamir\OneDrive\TSX-Trader  
 Step 2: Double-click start\_daemon.bat  
 (This runs: C:\Users\Aamir\miniforge3\python.exe telegram\_daemon.py)  
 Step 3: Verify the console shows: "Daemon listening for Telegram messages..."  
 Step 4: The Antigravity Agent will automatically send its first analysis at 10:00 AM ET via the cron schedule.

### 13.2 Logging a Trade via Telegram

You: I bought 50 CNQ at 60.50  
 Bot: Did you mean to BUY 50.0 shares of CNQ.TO @ \$60.50?  
 Reply YES to confirm or NO to cancel.  
 You: yes  
 Bot: Confirmed and Recorded: BUY 50.0 CNQ.TO @ \$60.50  
 CNQ.TO: 150.0 shares | Cash: \$10,681.00

### 13.3 Requesting On-Demand Analysis

You: analyze  
 Bot: Generating on-demand analysis. Please wait...  
 [Returns full SHORT or MEDIUM mode recommendation]

### 13.4 Asking a Market Question

You: What is the current RSI on CNQ?  
 Bot: Reading live market data to answer your question...  
 [Returns dossier-backed answer from Gemini]

### 13.5 Switching Trading Mode

You: change mode  
 Bot: Please reply with either Short Term Mode or Medium Term Mode.  
 You: Medium Term Mode  
 Bot: Operating mode successfully updated to: MEDIUM TERM.

### 13.6 Restoring the Agent After Session Loss

If the Antigravity agent session is lost (e.g., browser closed, session expired), the AGENT\_RESTORE\_INSTRUCTIONS.md file in the project root contains the exact cron expressions and prompt text needed to recreate all three scheduled tasks. Open a new session, share the file with the agent, and follow its instructions.

## 14. Troubleshooting Guide

### 14.1 Common Issues

#### **Bot is not responding to Telegram messages**

Cause 1: telegram\_daemon.py is not running. Check for the console window or re-run start\_daemon.bat.

Cause 2: TELEGRAM\_BOT\_TOKEN or TELEGRAM\_CHAT\_ID in .env is wrong. Verify with BotFather and your Telegram account.

Cause 3: chat\_id mismatch. Check the console for "Unauthorized access attempt" log lines.

#### **Analysis shows "Error fetching [TICKER]"**

Cause: Yahoo Finance is temporarily unavailable or the ticker symbol is invalid. Wait a few minutes and retry, or verify the ticker in config.json is a valid Yahoo Finance symbol.

#### **Cron-scheduled messages have stopped arriving**

Cause: The Antigravity agent session has been lost or the schedule tasks were cancelled. Open a new session and use AGENT\_RESTORE\_INSTRUCTIONS.md to recreate the three cron tasks.

#### **"Insufficient cash" error when logging a BUY**

Cause: The Cash\_Balance in trades.csv shows insufficient funds for the trade. Check the most recent Cash\_Balance value in trades.csv. If incorrect due to a manual edit, fix the last row's Cash\_Balance column.

#### **ValueError: Invalid action when logging a trade**

Cause: The Gemini intent-extraction returned an unexpected action value. This can happen if the user's message is ambiguous. Rephrase the trade clearly: "I bought 100 CNQ at 60.12".

#### **Telegram messages are being truncated**

Cause: A very long analysis is being sent by a code path that does not use chunking. The daemon's send\_chunked\_reply() handles this for /analyze and Q&A. The CLI send\_telegram.py also chunks to 4000 characters. If truncation occurs, check that the calling code is routing through one of these handlers.

#### **Advice history is empty on first run of the day**

This is expected behaviour. advice\_history.txt is populated throughout the day as scheduled and on-demand analyses are sent. By the second hourly analysis, the AI will have its first advice entry available as context.